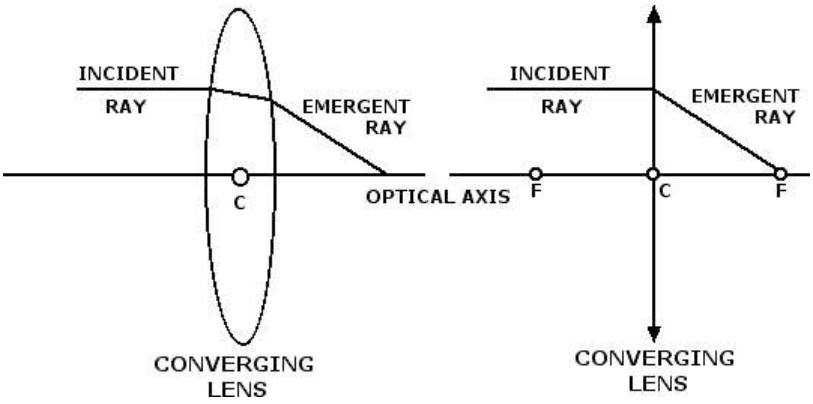
2 Converging Lenses

A **lens** is a transparent optical device placed between two spherical surfaces. Let us analyze what happens when a ray of light approaches the lens. There is a first refraction (from air to glass) when the ray of light enters the lens and a second refraction (from glass to air) when the ray of light emerges from the lens. In this section we will always consider lenses with negligible thickness: in this approximation (called *thin lens approximation*) the ray of light is deviated only once when it passes through the lens, as in the following picture, where the converging lens has been schematized by a double arrow:



Before going on, we have to introduce some other definitions: the **optical axis** is the symmetry axis of the lens. The symmetry axis intersects the lens in a point which is called **optical center** (C). In this section we will consider only **converging lenses**, i.e. those lenses where the emerging rays converge in a point called the **focal point** F of the lens. Each focal point is characterized by a distance from the optical center called **focal length**. Since every lens is formed by two mirrors joined together, every lens has two focal lengths.

As in the case of mirrors, also for converging lenses the image formation can be obtained by considering for each point of the object two particular rays:

1. the incident ray which is parallel to the optical axis: the emergent refracted ray passes through the focal point of the lens;

2. the incident ray passing through the optical center, which is not deviated by the lens.

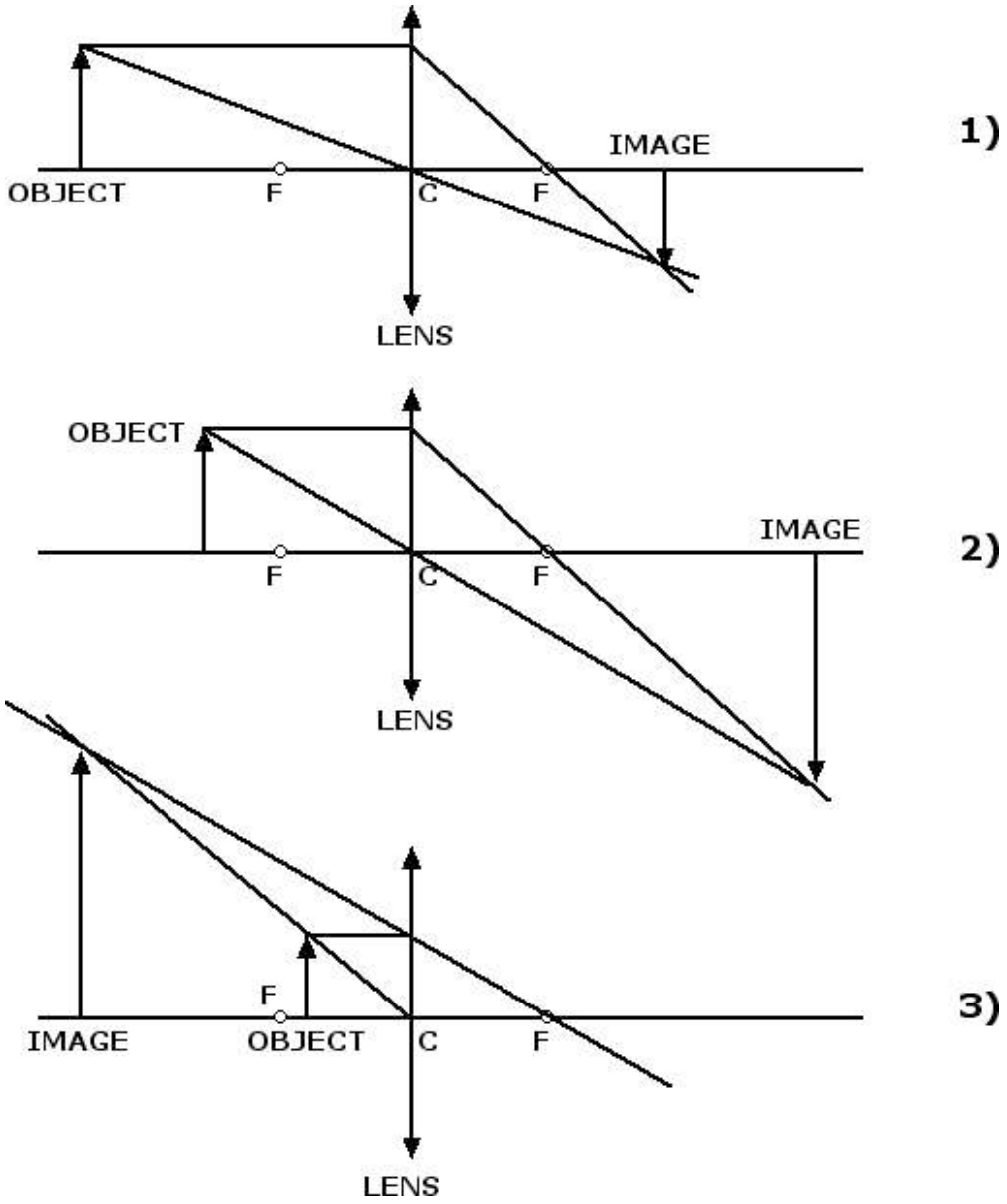
The images of a vertical line remains a vertical line. The features of the image depend on the distance p of the object from the lens and on the focal length f. In fact, using your pencil and your straightedge, you can find out that:

1. if the object is placed at a distance $p > 2f$ then the image turns out to be real, inverted and reduced in size;

2. if the distance p between the object and the lens satisfies $f < p < 2f$, then the image is real, inverted and enlarged;

3. if the object is placed between the focal point and the optical center $p < f$ then the image turns out to be virtual, upright and enlarged,

as it is clear from the following picture:



3 Diverging Lenses

A **diverging lens** has the peculiar feature that all the incident rays coming from directions which are parallel to the optical axis generate diverging emergent rays, i.e. the refracted rays never intersect. Nevertheless, if we extend the rays we find out that the extensions intersect in one of the two focal points of the lens.

As in the case of converging lenses, the image formation by a divergent lens can be done considering point by point the ray parallel to the optical axis (whose extension converges to